4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- **We will assume the beam is untapered (i.e. constant cross section) so that I_y, I_z, and I_{yz} do not vary in the x direction**

- **Therefore differentiating both sides of this formula with respect to x yields:**

$$\frac{\partial \sigma_x}{\partial x} = \frac{-y\left[\left(I_y\right)\frac{\partial M_z}{\partial x} + \left(I_{yz}\right)\frac{\partial M_y}{\partial x}\right] + z\left[\left(I_{yz}\right)\frac{\partial M_z}{\partial x} + \left(I_z\right)\frac{\partial M_y}{\partial x}\right]}{\left(I_y\right)\left(I_z\right) - \left(I_{yz}\right)^2}$$

- **Using the relations $V_y = -\frac{dM_z}{dx}$ and $V_z = \frac{dM_y}{dx}$ (derived at the end of Section 4.4 in the textbook) yields:**

$$\frac{\partial \sigma_x}{\partial x} = \frac{-y\left[\left(I_y\right)\left(-V_y\right) + \left(I_{yz}\right)\left(+V_z\right)\right] + z\left[\left(I_{yz}\right)\left(-V_y\right) + \left(I_z\right)\left(+V_z\right)\right]}{\left(I_y\right)\left(I_z\right) - \left(I_{yz}\right)^2}$$

4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- **Recall the boxed equation from Mar 13 →** $$\boxed{(q_{out}) - (q_{in}) = \int_{A'} \frac{\partial \sigma_x}{\partial x} dA}$$

- **Inserting our expression for $\frac{d\sigma_x}{dx}$ into this equation yields:**

$$(q_{out}) - (q_{in}) = \int_{A'} \left\{ \frac{-y\left[(I_y)(-V_y) + (I_{yz})(+V_z)\right] + z\left[(I_{yz})(-V_y) + (I_z)(+V_z)\right]}{(I_y)(I_z) - (I_{yz})^2} \right\} dA$$

- **Rearranging gives:**

$$(q_{out}) - (q_{in}) = \frac{(I_y)V_y - (I_{yz})V_z}{(I_y)(I_z) - (I_{yz})^2} \int_{A'} y\, dA + \frac{-(I_{yz})V_y + (I_z)V_z}{(I_y)(I_z) - (I_{yz})^2} \int_{A'} z\, dA$$

- **Recall that $\int_A ydA = Q_z = \bar{y}A$ and $\int_A zdA = Q_y = \bar{z}A$ (the first moments of area about the *z* and *y* axes, respectively)**

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- **Inserting these expressions into the above equation yields the General Shear Flow Formula:**

$$\boxed{q_{out} - q_{in} = \frac{\left(I_y V_y - I_{yz} V_z\right)\bar{y}' A' + \left(-I_{yz} V_y + I_z V_z\right)\bar{z}' A'}{I_y I_z - I_{yz}^{2}}}$$

- **We can see how the equation simplifies in certain circumstances:**
  - When $V_y = 0$, $q_{out} - q_{in} = \frac{(\bar{z} I_z - \bar{y} I_{yz}) V_z A'}{I_y I_z - I_{yz}^2}$
  - When $V_z = 0$, $q_{out} - q_{in} = \frac{(\bar{y} I_y - \bar{z} I_{yz}) V_y A'}{I_y I_z - I_{yz}^2}$
  - When $I_{yz} = 0$, $q_{out} - q_{in} = \frac{\bar{y} I_y V_y A + \bar{z} I_z V_z A'}{I_y I_z}$
  - When $V_y = 0$ and $I_{yz} = 0$, $q_{out} - q_{in} = \frac{V_z Q_y}{I_y}$
  - When $V_z = 0$ and $I_{yz} = 0$, $q_{out} - q_{in} = \frac{V_y Q_z}{I_z}$

This leads to the $\tau = \frac{VQ}{Ib}$ equation we are most familiar with!